

S.3 Geography · Topic 2: Living with Natural Hazards
6-Mark Essay Question Bank
Suggested answers in point form — for memorising

How to use: each essay is worth 6 marks. Suggested answers give MORE than 6 points so you can choose the strongest ones. In the exam, write in full sentences, use the geography terms in bold, and aim for 6–8 developed points. Memory aids are given to help you recall the lists.

Essay 1 (6 marks)

Explain why losses caused by earthquakes of SIMILAR magnitude vary between countries of different levels of economic development.

Opening definition (write first):

Countries with a higher level of economic development are more developed countries (MDCs); those with a lower level are less developed countries (LDCs).

Suggested points (1 mark each):

- MDCs have more capital and higher technology to reduce potential losses.
- MDCs spend more on research & development, so earthquakes are better studied and more precautions / remedial measures exist.
- MDCs can afford effective monitoring and warning systems (e.g. J-Alert in Japan) → early evacuation.
- MDCs enforce building laws and earthquake-proof designs (e.g. dampers) → buildings do not collapse.
- MDCs have well-trained, well-equipped rescue teams → fewer deaths.
- People in MDCs have higher education and hold regular drills → higher awareness and preparedness.
- MDCs can take out insurance and provide disaster aid to rebuild.
- LDC buildings are made of stone, brick or wood → cannot withstand shaking or fire.
- LDCs have poorly-developed transport networks → rescue work is hindered.
- LDCs lack money for warning systems; lower education and corruption increase losses.

Conclusion (write last):

In conclusion, although the earthquakes are of similar magnitude, MDCs suffer smaller losses than LDCs because they have the capital, technology, stronger buildings and better-prepared people needed to reduce the impact — so the level of economic development is the key factor.

Memory aid: MRBPEG — Money · Roads · Buildings · People (density) · Education · Government.

Exam tip: Always start by defining MDC and LDC, then contrast them point-by-point.

Essay 2 (6 marks)

An earthquake of high magnitude struck, yet casualties and losses were minimal. Explain why.

Opening definition (write first):

The level of destruction depends on the characteristics of the hazard AND of the affected area, not on magnitude alone.

Suggested points (1 mark each):

- The focus was deep, so seismic waves weakened before reaching the surface.
- The epicentre / nearest town was far away (long horizontal distance) → weaker shaking.
- Low population density (or uninhabited area) → few people exposed.
- An early earthquake warning was issued → people could evacuate.
- It occurred in the daytime, when people have higher awareness.
- It happened in an MDC with capital for rescue teams, warning systems and education.
- Buildings used earthquake-proof designs, so few structures collapsed.
- People were well prepared through drills and education.

Conclusion (write last):

In conclusion, a high magnitude alone does not mean heavy losses; because the focus was deep, the area thinly populated, and effective warning and preparation were in place, the casualties and damage remained minimal.

Memory aid: Think: deep focus · far away · few people · warned · daytime · MDC measures.

Exam tip: Link each point to 'so fewer deaths / less damage' to show cause and effect.

Essay 3 (6 marks)

Explain how technology and other measures can reduce the damage caused by earthquakes.

Opening definition (write first):

There are six general types of measure used to cope with tectonic hazards.

Suggested points (1 mark each):

- Monitoring & warning systems (seismographs, J-Alert) raise awareness and give time to evacuate.
- Hazard-proof structures: earthquake-proof designs and dampers prevent buildings collapsing.
- Land-use planning: hazard-zone maps control development and keep open areas as shelters.
- Education & drills increase public awareness so people know how to respond.
- Rescue systems: trained teams, life detectors and helicopters reduce the number of deaths.
- Post-hazard finance: insurance and disaster aid help rebuild society.
- Communication and satellite/radar monitoring help detect hazards early.

Conclusion (write last):

In conclusion, no single measure is enough; combining monitoring and warning, hazard-proof building, land-use planning, education, rescue systems and post-hazard finance gives the most effective protection against earthquake damage.

Memory aid: M-B-P-E-R-F — Monitor · Build · Plan · Educate · Rescue · Fund.

Exam tip: Name the measure, then say what it DOES. Six measures = full marks.

Essay 4 (6 marks)

With reference to plate movement, explain how earthquakes are formed at a **CONVERGENT** plate boundary.

Opening definition (write first):

An earthquake is the sudden shaking of rocks in the Earth's crust caused by plate movement.

Suggested points (1 mark each):

- Convection currents in the mantle converge and drag the two plates towards each other.
- The plates collide along the convergent boundary under compressional force.
- Stress is applied to the rocks, which are held back by friction.
- Pressure builds up within the rocks and the stress increases.
- When the stress exceeds the strength of the rock, the rock breaks and displaces.
- The stored energy is released and transmitted as seismic waves.
- The seismic waves move out in all directions and shake the ground → an earthquake forms.

Conclusion (write last):

In conclusion, the collision of two plates under compressional force at a convergent boundary builds up stress until the rock breaks and releases energy as seismic waves, which is how the earthquake is formed.

Memory aid: 6 steps: move → stress/friction → pressure builds → rock breaks → energy released → waves shake.

Exam tip: For a divergent boundary, swap to 'convection diverges → plates move apart → tensional force'.

Essay 5 (6 marks)

Explain why a country such as Japan is prone to earthquakes.

Opening definition (write first):

Japan lies on the Pacific Ring of Fire (Circum-Pacific Belt), where several plates meet.

Suggested points (1 mark each):

- Japan sits at or near the boundary of several plates (e.g. Pacific, Philippine, Eurasian, North American).
- Mantle convection currents keep these plates moving (about 5–10 cm/year).
- At convergent boundaries the plates collide under compressional force.
- Stress builds up in the rocks until it exceeds their strength and they break.
- Energy is released as seismic waves, producing frequent earthquakes.
- Being on the Ring of Fire means many active boundaries → high earthquake frequency.
- Earthquakes here can also trigger tsunamis along the coast.

Conclusion (write last):

In conclusion, because Japan lies on the Pacific Ring of Fire where several moving plates meet, stress is constantly built up and released along the boundaries, making the country highly prone to earthquakes.

Memory aid: Near boundary → convection moves plates → collide → break → frequent quakes.

Exam tip: Name real plates and the Ring of Fire to score the place-specific marks.

Essay 6 (6 marks)

Explain the factors that affect the LEVEL OF DESTRUCTION caused by an earthquake.

Opening definition (write first):

Destruction depends on the characteristics of the affected area and of the hazard itself.

Suggested points (1 mark each):

- Population density — higher density means more people are exposed (more casualties).
- Level of economic development — more infrastructure means greater property loss.
- Preparedness / awareness — higher awareness reduces losses.
- Magnitude — higher magnitude releases more energy → more destruction.
- Focal depth — a shallow focus loses less energy → more destructive.
- Distance from source — closer to the epicentre → stronger shaking → more destruction.
- Time of occurrence — at midnight people have low awareness → more deaths.
- Duration and frequency — longer/less frequent hazards leave people less able to cope.

Conclusion (write last):

In conclusion, the level of destruction is decided not by one factor but by the combined characteristics of the hazard (magnitude, focal depth, distance, time) and of the affected area (population density, development and preparedness).

Memory aid: Two groups: AREA (density, development, preparedness) + HAZARD (magnitude, depth, distance, time).

Exam tip: Group your points under 'the area' and 'the hazard' to stay organised.

Essay 7 (6 marks)

Describe the short-term and long-term effects (impacts) of a major earthquake.

Opening definition (write first):

An earthquake brings both immediate (short-term) effects and lasting (long-term) effects on people, property and the economy.

Suggested points (1 mark each):

- Short-term: buildings and houses are destroyed.
- Short-term: transport links (roads, railways, bridges) are damaged, hindering rescue.
- Short-term: public utilities (water, gas, electricity) are cut off.
- Short-term: gas leakage leads to fire; landslides, tsunami or flooding may follow.
- Short-term: spread of disease and the danger of aftershocks.
- Long-term: economic development of the region is obstructed for years.
- Long-term: victims suffer bad psychological effects (trauma, anxiety).
- Long-term: social instability such as riots and robbery may occur.

Conclusion (write last):

In conclusion, a major earthquake causes immediate destruction of buildings, transport and utilities, while its long-term effects — economic setback, psychological harm and social instability — can last for years.

Memory aid: Short-term = happens at once (collapse, fire, utilities). Long-term = lasting (economy, trauma, social).

Exam tip: Split your answer clearly into 'short-term' and 'long-term' to organise the marks.